Selection of Software Process Model for the Project

Project Title

Campus Library Resource Search Engine

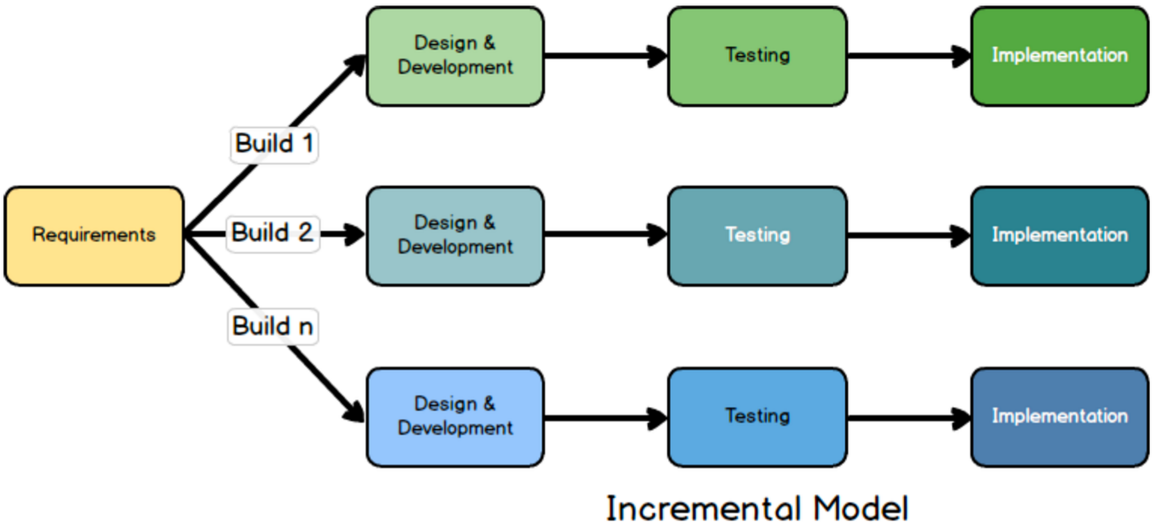
Introduction

Software development models help in organizing the development process of a software system. Choosing an appropriate development model is important because it affects project efficiency, team coordination, and final product quality.

For the development of the **Campus Library Resource Search Engine**, the **Incremental Process Model** is selected as the most suitable development approach.

The project is being developed by a **team of four members** with a **deadline of one week**, therefore a model that supports modular development, parallel work, and quick delivery is required.

Incremental Process Model



The Incremental Process Model is a software development approach in which the system is developed and delivered in small parts called **increments**. Each increment includes the phases of requirement analysis, design, implementation, and testing.

In this model, the system is built step-by-step, and each increment adds new functionality to the existing system until the complete software is developed.

Instead of building the entire system at once, the project is divided into smaller modules, and each module is developed separately.

Reasons for Selecting the Incremental Model

1. Short Development Time

The project must be completed within **one week**. The Incremental Model allows the development team to build the system in smaller parts, making it easier to manage and complete within the limited time available.

Each increment can be developed quickly, tested, and integrated into the main system.

2. Suitable for Team Development

The project team consists of **four members**. The Incremental Model allows tasks to be divided among team members so that multiple modules can be developed at the same time.

For example, different members can work on different modules such as the user interface, backend service, database management, and admin panel.

This improves productivity and ensures that the project progresses faster.

3. Modular System Design

The Campus Library Resource Search Engine naturally consists of multiple modules, including:

- Search Interface
- Backend API
- Database
- Search Engine Integration
- Admin Management Panel

The Incremental Model supports modular development, allowing each module to be developed independently and later integrated into the final system.

4. Early Working Prototype

One of the advantages of the Incremental Model is that a working version of the system can be produced early in the development process.

Even if all features are not completed, the system can still perform basic functions such as searching resources and displaying results.

This helps in verifying whether the system works correctly and allows improvements to be made during development.

5. Easy Testing and Debugging

Testing is performed after each increment is completed. This makes it easier to identify and fix errors in smaller modules rather than testing the entire system at once.

As a result, debugging becomes simpler and the quality of the software improves.

Development Plan Using Incremental Model

The project development can be divided into the following increments:

Increment 1 – User Interface Development

In this phase, the basic search interface will be developed. The user interface will include a search bar and a results display page.

Increment 2 – Backend Development

The backend services will be developed to handle user requests and communicate with the database and search engine.

Increment 3 – Database and Search Engine Integration

In this phase, the database will store resource metadata such as title, author, and publication year. The search engine will index the resources to enable fast searching.

Increment 4 – Admin Panel Development

The administrative panel will allow administrators to add, update, or remove resources from the system.

Increment 5 – System Integration and Testing

Finally, all modules will be integrated and tested to ensure that the system functions correctly. Testing methods such as white-box testing and black-box testing will be used.

Conclusion

The Incremental Process Model is the most appropriate development model for this project because it supports modular development, allows parallel work among team members, and enables faster delivery of functional components.